

The same compiled rule, growing its guards: a violation is a non-empty query result.

```
SELECT param_name, step,  
       COUNT(DISTINCT cksum) AS n  
FROM trace  
GROUP BY param_name, step  
HAVING n > 1;
```

*no topology filter:
every TP-sharded tensor
becomes a violation*



(a) π_{schema} only (83.3 FP/1M)

```
SELECT param_name, step,  
       COUNT(DISTINCT cksum) AS n  
FROM trace  
WHERE tensor_model_  
       parallel = FALSE  
GROUP BY param_name, step  
HAVING n > 1;
```

*guard in the query:
sharded tensors filtered
before the check*

*still fires at step 0,
before weight sync*



(b) + π_{topo} (shards excluded)

```
SELECT param_name, step,  
       COUNT(DISTINCT cksum) AS n  
FROM trace  
WHERE tensor_model_  
       parallel = FALSE  
AND stage =  
      'after_optimizer'  
GROUP BY param_name, step  
HAVING n > 1;
```

*valid only after
the optimizer step:
clean run silent,
bug DETECTED*



(c) verified rule (25.8 FP/1M, 0/17 FP)